

HTML HW6

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Consider the case where $\mathbf{z}_n^T = [1, \mathbf{x}_n^T]$ is fed into D_1

$$\begin{aligned} \min \quad & \frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m y_n y_m \mathbf{z}_n^T \mathbf{z}_m - \sum_{n=1}^N \alpha_n \\ \text{Subject to} \quad & \sum_{n=1}^N y_n \alpha_n = 0, \quad 0 \leq \alpha_n \leq C \quad \forall n \in \{1, 2, \dots, N\} \end{aligned}$$

Notice that

$$\mathbf{z}_n^T \mathbf{z}_m = \begin{bmatrix} 1 & \mathbf{x}_n^T \end{bmatrix} \begin{bmatrix} 1 \\ \mathbf{x}_m \end{bmatrix} = 1 + \mathbf{x}_n^T \mathbf{x}_m$$

So D_1 becomes

$$\begin{aligned} \min \quad & \frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m y_n y_m \mathbf{x}_n^T \mathbf{x}_m + \frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m y_n y_m - \sum_{n=1}^N \alpha_n \\ \text{Subject to} \quad & \sum_{n=1}^N y_n \alpha_n = 0, \quad 0 \leq \alpha_n \leq C \quad \forall n \in \{1, 2, \dots, N\} \end{aligned}$$

The only difference between this and the original form of the problem is the $\frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m y_n y_m$ term

$$\frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m y_n y_m = \sum_{m=1}^N \left(\alpha_m y_m \sum_{n=1}^N \alpha_n y_n \right)$$

Since we have the constraint $\sum_{n=1}^N y_n \alpha_n = 0$, we know that the term above is 0, and the two optimization problem in fact solve for the same α

Let the solution to the original problem be $(b, \mathbf{w}, \alpha)$

Hence, $\alpha^* = \alpha$

From the lecture slides, we know that $\mathbf{w} = \sum_{n=1}^N \alpha_n y_n \mathbf{z}_n$, so we have

$$\mathbf{w} = \sum_{n=1}^N \alpha_n y_n \mathbf{x}_n, \quad \tilde{\mathbf{w}}^* = \sum_{n=1}^N \alpha_n y_n [1, \mathbf{x}_n^T]^T$$

So

$$\begin{aligned}\tilde{\mathbf{w}}^* &= \left[\sum_{n=1}^N \alpha_n y_n, \mathbf{w} \right]^T \\ &= [0 \ \mathbf{w}]^T \quad (\text{Because we have the constraint } \sum_{n=1}^N \alpha_n y_n = 0)\end{aligned}$$

Hence $\mathbf{w}^* = \mathbf{w}$

Also from the lecture slides, $b = y_s - \mathbf{w}^T \mathbf{z}_s$, where $\mathbf{z}_s$ is some free support vector.

So, with some $0 < \alpha_s < C$

$$\begin{aligned}b^* &= y_s - \tilde{\mathbf{w}}^{*T} \mathbf{z}_s \\ &= y_s - [0 \ \mathbf{w}] [1 \ \mathbf{x}_s]^T \\ &= y_s - \mathbf{w}^T \mathbf{x}_s\end{aligned}$$

And

$$b = y_s - \mathbf{w}^T \mathbf{x}_s$$

Hence we also see that $b^* = b$

From the analysis above, we see that $(b^*, \mathbf{w}^*, \alpha^*) = (b, \mathbf{w}, \alpha)$, and hence $(b^*, \mathbf{w}^*, \alpha^*) = (b, \mathbf{w}, \alpha)$ is an optimal solution to the original problem.

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Original problem:

$$\begin{aligned}
 (P) \quad & \min_{\mathbf{w}, b, \xi} \quad \frac{1}{2} \mathbf{w}^T \mathbf{w} + C \sum_{n=1}^N \xi_n \\
 \text{Subject to} \quad & y_n(\mathbf{w}^T \Phi(\mathbf{x}_n) + b) \geq 1 - \xi_n, \text{ for } n = 1, 2, \dots, N \\
 & \xi_n \geq 0, \text{ for } n = 1, 2, \dots, N \\
 & y_0(\mathbf{w}^T \Phi(\mathbf{w}_0) + b) \geq 1
 \end{aligned}$$

Define $\mathbf{z}_n = \Phi(\mathbf{w}_n)$

This is equivalent to

$$\min_{b, \mathbf{w}, \xi} \left(\max_{0 \leq \alpha_n, 0 \leq \beta_n} \frac{1}{2} \mathbf{w}^T \mathbf{w} + C \sum_{n=1}^N \xi_n + \sum_{n=1}^N \alpha_n (1 - \xi_n - y_n(\mathbf{w}_n^T \mathbf{z}_n) + b) + \alpha_0 (1 - y_0(\mathbf{w}_0^T \mathbf{z}_0) + b) + \sum_{n=1}^N \beta_n (-\xi_n) \right)$$

by the same arguments as introduced in class.

Now the (strong) duality of QP allows us to transform this into its Lagrange dual

$$\max_{0 \leq \alpha_n, 0 \leq \beta_n} \left(\min_{b, \mathbf{w}, \xi} \frac{1}{2} \mathbf{w}^T \mathbf{w} + C \sum_{n=1}^N \xi_n + \sum_{n=1}^N \alpha_n (1 - \xi_n - y_n(\mathbf{w}_n^T \mathbf{z}_n) + b) + \alpha_0 (1 - y_0(\mathbf{w}_0^T \mathbf{z}_0) + b) + \sum_{n=1}^N \beta_n (-\xi_n) \right)$$

First we deal with the inner problem

$$\begin{aligned}
 & \min_{b, \mathbf{w}, \xi} \frac{1}{2} \mathbf{w}^T \mathbf{w} + C \sum_{n=1}^N \xi_n + \sum_{n=1}^N \alpha_n (1 - \xi_n - y_n(\mathbf{w}_n^T \mathbf{z}_n) + b) + \alpha_0 (1 - y_0(\mathbf{w}_0^T \mathbf{z}_0) + b) + \sum_{n=1}^N \beta_n (-\xi_n) \\
 & = \min_{b, \mathbf{w}, \xi} \mathcal{L}(b, \mathbf{w}, \alpha)
 \end{aligned}$$

At optimal we have

$$\frac{\partial \mathcal{L}}{\partial b} = 0 = - \sum_{n=0}^N \alpha_n y_n$$

Hence there's no loss of optimality if solving with one extra constraint $\sum_{n=0}^N \alpha_n y_n = 0$.

Solving under this constraint, we see that b is removed from the problem, and we now have the dual problem in the form

$$\max_{0 \leq \alpha_n, 0 \leq \beta_n, \sum \alpha_n y_n = 0} \left(\min_{b, \mathbf{w}, \xi} \frac{1}{2} \mathbf{w}^T \mathbf{w} + C \sum_{n=1}^N \xi_n + \sum_{n=1}^N \alpha_n (1 - \xi_n - y_n(\mathbf{w}_n^T \mathbf{z}_n)) + \alpha_0 (1 - y_0(\mathbf{w}_0^T \mathbf{z}_0)) + \sum_{n=1}^N \beta_n (-\xi_n) \right)$$

Moreover, the optimality of the inner problem implies that

$$\frac{\partial \mathcal{L}}{\partial \xi_i} = 0 = C - \alpha_n - \beta_n \quad \text{for } n = 1, 2, \dots, N$$

Hence there's also no loss of optimality if solving under the constraints $\beta_n = C - \alpha_n$, $n = 1, 2, \dots, N$ and $0 \leq \alpha_n \leq C$

Solving under these new constraints, ξ is also removed from the problem, and the Lagrange dual problem now becomes

$$\max_{\sum \alpha_n y_n = 0, 0 \leq \alpha_1, \alpha_2, \dots, \alpha_N \leq C, 0 \leq \alpha_0, \beta_1 = C - \alpha_1, \dots, \beta_N = C - \alpha_N} \left(\min_{b, \mathbf{w}, \xi} \frac{1}{2} \mathbf{w}^T \mathbf{w} + \sum_{n=0}^N \alpha_n (1 - y_n (\mathbf{w}_n^T \mathbf{z}_n)) \right)$$

Lastly, the optimality of the inner problem provides us with the final condition

$$\frac{\partial \mathcal{L}}{\partial w_i} = 0 = w_i - \sum_{n=0}^N \alpha_n y_n z_{n,i}$$

Hence there is no loss of optimality solving with constraint $\mathbf{w} = \sum_{n=0}^N \alpha_n y_n \mathbf{z}_n$

Finally, the Lagrange dual problem becomes

$$\begin{aligned} & \max_{\sum \alpha_n y_n = 0, 0 \leq \alpha_1, \dots, \alpha_N \leq C, 0 \leq \alpha_0, \beta_1 = C - \alpha_1, \dots, \beta_N = C - \alpha_N, \mathbf{w} = \sum \alpha_n y_n \mathbf{z}_n} \left(\min_{b, \mathbf{w}, \xi} \frac{1}{2} \mathbf{w}^T \mathbf{w} + \sum_{n=0}^N \alpha_n - \mathbf{w}^T \mathbf{w} \right) \\ \Leftrightarrow & \max_{\sum \alpha_n y_n = 0, 0 \leq \alpha_1, \dots, \alpha_N \leq C, 0 \leq \alpha_0, \beta_1 = C - \alpha_1, \dots, \beta_N = C - \alpha_N, \mathbf{w} = \sum \alpha_n y_n \mathbf{z}_n} \left(\min_{b, \mathbf{w}, \xi} -\frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m y_n y_m \mathbf{z}_n^T \mathbf{z}_m + \sum_{n=0}^N \alpha_n \right) \\ \Leftrightarrow & \min_{\sum \alpha_n y_n = 0, 0 \leq \alpha_1, \dots, \alpha_N \leq C, 0 \leq \alpha_0, \beta_1 = C - \alpha_1, \dots, \beta_N = C - \alpha_N, \mathbf{w} = \sum \alpha_n y_n \mathbf{z}_n} \left(\frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m y_n y_m \mathbf{z}_n^T \mathbf{z}_m - \sum_{n=0}^N \alpha_n \right) \end{aligned}$$

Now we have the complete dual problem

$$\begin{aligned} (D) \quad & \min_{\alpha} \left(\frac{1}{2} \sum_{n=0}^N \sum_{m=0}^N \alpha_n \alpha_m y_n y_m \mathbf{z}_n^T \mathbf{z}_m - \sum_{n=0}^N \alpha_n \right) \\ \text{Subject to} \quad & \sum_{n=0}^N \alpha_n y_n = 0 \\ & 0 \leq \alpha_n \leq C \quad \text{for } n = 1, 2, \dots, N \\ & 0 \leq \alpha_0 \\ \text{Implicitly } & \mathbf{w} = \sum_{n=0}^N \alpha_n y_n \mathbf{z}_n \\ & \beta_n = C - \alpha \quad \text{for } n = 1, 2, \dots, N \end{aligned}$$

Finally, we see that this is a QP problem

$$\begin{aligned} \min_{\alpha} \quad & \frac{1}{2} \alpha^T \mathbf{Q} \alpha + \mathbf{p}^T \alpha \\ \text{Subject to} \quad & \mathbf{a}_i^T \alpha \geq c_i \end{aligned}$$

With the parameters defined as follows

$$\begin{aligned} \mathbf{Q}_{ij} = y_i y_j \mathbf{z}_i^T \mathbf{z}_j &= \begin{cases} -\Phi(\mathbf{x}_i)^T \Phi \mathbf{x}_j & 0 = i \neq j \text{ or } 0 = j \neq i \\ \Phi(\mathbf{x}_i)^T \Phi \mathbf{x}_j & \text{else} \end{cases} \\ \mathbf{p} = -\mathbf{1}_{N+1} &= \underbrace{[-1, -1, \dots, -1]}_{N+1} \\ \mathbf{a}_{\geq} = \mathbf{y} &= [-1, \underbrace{1, 1, \dots, 1}_N]^T \quad c_{\geq} = 0 \quad \left(\sum_{n=0}^N \alpha_n y_n \geq 0 \right) \\ \mathbf{a}_{\leq} = -\mathbf{y} &= [1, \underbrace{-1, -1, \dots, -1}_N]^T \quad c_{\leq} = 0 \quad \left(\sum_{n=0}^N \alpha_n y_n \leq 0 \right) \\ \mathbf{a}_{n \geq} &= [\underbrace{0, \dots, 0}_n, 1, \underbrace{0, \dots, 0}_{N-n-1}] \quad c_{n \geq} = 0 \quad \text{for } n = 0, 1, 2, \dots, N \quad (\alpha_n \geq 0) \\ \mathbf{a}_{n \leq} &= [\underbrace{0, \dots, 0}_n, -1, \underbrace{0, \dots, 0}_{N-n-1}] \quad c_{n \leq} = -C \quad \text{for } n = 1, 2, \dots, N \quad (\alpha_n \leq C) \end{aligned}$$

With optimal α solved with QP solver, we can in turn solve $\mathbf{w}$ and b with the KKT conditions.

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$$\hat{h}(\mathbf{x}) = \text{sign} \left(\sum_{n=1}^N y_n K(\mathbf{x}_n, \mathbf{x}) \right)$$

Next,

$$\begin{aligned} \hat{h}(\mathbf{x}_n) &= \text{sign} \left(y_n K(\mathbf{x}_n, \mathbf{x}_n) + \sum_{m \neq n} y_m K(\mathbf{x}_n, \mathbf{x}_m) \right) \\ &= \text{sign} \left(y_n + \sum_{m \neq n} y_m \exp(-\gamma \|\mathbf{x}_n - \mathbf{x}_m\|^2) \right) \end{aligned}$$

Since the distance between any two different $\mathbf{x}_n, \mathbf{x}_m$ in the $\mathcal{X}$ -space is greater than or equal to ϵ we have

$$\|\mathbf{x}_n - \mathbf{x}_m\|^2 \geq \epsilon^2 \quad \forall n \neq m$$

Also, $\gamma > \frac{\ln(N-1)}{\epsilon^2}$, so

$$\begin{aligned} \gamma \|\mathbf{x}_n - \mathbf{x}_m\|^2 &> \frac{\ln(N-1)}{\epsilon^2} \epsilon^2 = \ln(N-1) \\ \Rightarrow \exp(-\gamma \|\mathbf{x}_n - \mathbf{x}_m\|^2) &< \frac{1}{N-1} \\ \Rightarrow |y_m \exp(-\gamma \|\mathbf{x}_n - \mathbf{x}_m\|^2)| &< \frac{|y_m|}{N-1} = \frac{1}{N-1} \end{aligned}$$

Hence

$$\begin{aligned} \sum_{m \neq n} -|y_m \exp(-\gamma \|\mathbf{x}_n - \mathbf{x}_m\|^2)| &< \sum_{m \neq n} y_m \exp(-\gamma \|\mathbf{x}_n - \mathbf{x}_m\|^2) < \sum_{m \neq n} |y_m \exp(-\gamma \|\mathbf{x}_n - \mathbf{x}_m\|^2)| \\ \Rightarrow -1 &< \sum_{m \neq n} y_m \exp(-\gamma \|\mathbf{x}_n - \mathbf{x}_m\|^2) < 1 \end{aligned}$$

Therefore

$$y_n - 1 < y_n + \sum_{m \neq n} y_m \exp(-\gamma \|\mathbf{x}_n - \mathbf{x}_m\|^2) < y_n + 1$$

and

$$\text{sign} \left(y_n + \sum_{m \neq n} y_m \exp(-\gamma \|\mathbf{x}_n - \mathbf{x}_m\|^2) \right) = \text{sign}(y_n) \quad (\text{Because } |y_n| = 1)$$

So we have $\hat{h}(\mathbf{x}_n) = y_n \quad \forall \mathbf{x}_n \in \mathcal{X}$, and therefore $E_{\text{in}}(\hat{h}) = 0$

Q.E.D.

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For any $x \in \mathbb{R}$, define the vector $\mathbf{x} = [\cos x \ \sin x]^T$. Notice that $\|\mathbf{x}\| = 1$.
Then

$$\begin{aligned} K(x, x') &= \exp(2 \cos(x - x') - 2) \\ &= \exp(2(\cos x \cos x' + \sin x \sin x') - 2) \\ &= \exp(2(\mathbf{x}^T \mathbf{x}') - 2) \\ &= \exp(2(\mathbf{x}^T \mathbf{x}') - \|\mathbf{x}\|^2 - \|\mathbf{x}'\|^2) \\ &= \exp(-\|\mathbf{x} - \mathbf{x}'\|^2) \end{aligned}$$

Notice that this has the same form as the Gaussian kernel.

Now, consider Mercer's condition.

The $N \times N$ matrix $\mathbf{K}$ is defined by

$$K_{ij} = K(\mathbf{x}_i, \mathbf{x}_j)$$

Mercer's condition states that $\mathbf{K}$ is positive semi-definite iff K is a valid kernel. Also, we're taught in class that the Gaussian kernel is a valid kernel.

Therefore, since the matrix $\mathbf{K}$ has the same form as in the scenario when using the Gaussian kernel on N vectors with norm 1 (this constraint should obviously not affect the validity of the Gaussian kernel), $\mathbf{K}$ is positive semi-definite, and we know that K is indeed a valid kernel.

Q.E.D.

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Define $\phi_i(\mathbf{x})$ as

$$\phi(\mathbf{x}) = [g_{i,\theta_1}(\mathbf{x}), g_{i,\theta_2}(\mathbf{x}), \dots, g_{i,\theta_k}(\mathbf{x})]^T$$

We can see that

$$K_{ds}(\mathbf{x}, \mathbf{x}') = (\Phi_{ds}(\mathbf{x}))^T (\Phi_{ds}(\mathbf{x}')) = \sum_{i=1}^d (\phi_i(\mathbf{x}))^T (\phi_i(\mathbf{x}'))$$

Now we examine $(\phi_i(\mathbf{x}))^T (\phi_i(\mathbf{x}'))$

$$\phi_i(\mathbf{x}) = [\llbracket x_i > L + 0.5 \rrbracket, \llbracket x_i > L + 1.5 \rrbracket, \dots, \llbracket x_i > R - 0.5 \rrbracket]$$

Together with the fact that x_i is an integer in $[L, R]$, we have

$$\phi_i(\mathbf{x}) = [\underbrace{1, 1, \dots, 1}_{x_i - L \text{ 1s}}, \underbrace{0, 0, \dots, 0}_{d + L - x_i \text{ 0s}}]$$

It follows that

$$(\phi_i(\mathbf{x}))^T (\phi_i(\mathbf{x}')) = \min(x_i - L, x'_i - L) = \min(x_i, x'_i) - L$$

Therefore, we can conclude that

$$\begin{aligned} K_{ds}(\mathbf{x}, \mathbf{x}') &= (\Phi_{ds}(\mathbf{x}))^T (\Phi_{ds}(\mathbf{x}')) = \sum_{i=1}^d (\phi_i(\mathbf{x}))^T (\phi_i(\mathbf{x}')) \\ &= \sum_{i=1}^d (\min(x_i, x'_i) - L) \\ &= \sum_{i=1}^d \min(x_i, x'_i) - dL \end{aligned}$$

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Table:

$\begin{array}{c} \diagup \\ C \\ \diagdown \\ Q \end{array}$	0.1	1	10
2	505	505	505
3	547	547	547
4	575	575	575

The number in each field of the table is the corresponding number of support vectors.

$(C, Q) = (0.1, 2), (1, 2), (10, 2)$ results in the smallest number of support vectors. 505 precisely.

Findings:

From the table we can easily see that the number of SVs increase with Q , which is the degree of the polynomial transform used, and is not affected by C .

These two phenomena can be addressed separately.

First, a possible explanation to the positive relation between the number of SVs and Q is that, as introduced in class, SVs are the data vectors on the boundary of the hyperplane, and locates the hyperplane when solving SVM problems. Since a higher value of Q results in a model with higher degree of freedom, and hence hyperplanes with more complex shape, it is reasonable that it takes more support vectors to locate the optimal hyperplane.

Second, the reason that the number of SVs is not affected by C might be that the model used in this problem can all separate the data vectors fairly well. That is, most data vectors are correctly classified, and there are few cases of violation. As a result, the penalty for violation, C , does not affect the training result, and hence the number of SVs much.

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Table:

$\gamma \backslash C$	0.1	1	10
0.1	0.04127	0.01817	0.01779
1	0.090780	0.0090780	0.0089836
10	0.090793	0.0090793	0.0089819

The number in each field of the table is the corresponding value of $\frac{1}{\|w\|}$

$(Q, \gamma) = (0.1, 10)$ results in the largest margin. 0.090793 approximately

Findings:

From the table we can see that the same γ , the margin is smaller with larger C . Also, for the same C , the margin is larger with larger γ .

These two phenomena can be explained separately.

First, larger C means that violation is penalized more. Also, we know that the soft-margin SVM finds a balance between maximizing the margin and minimizing violation. Therefore, a higher penalty for violation means that this balance leans more towards minimizing violation, and the priority for maximizing the margin becomes lower. So it is reasonable that compared with a model with small C , one with larger C results in a smaller margin.

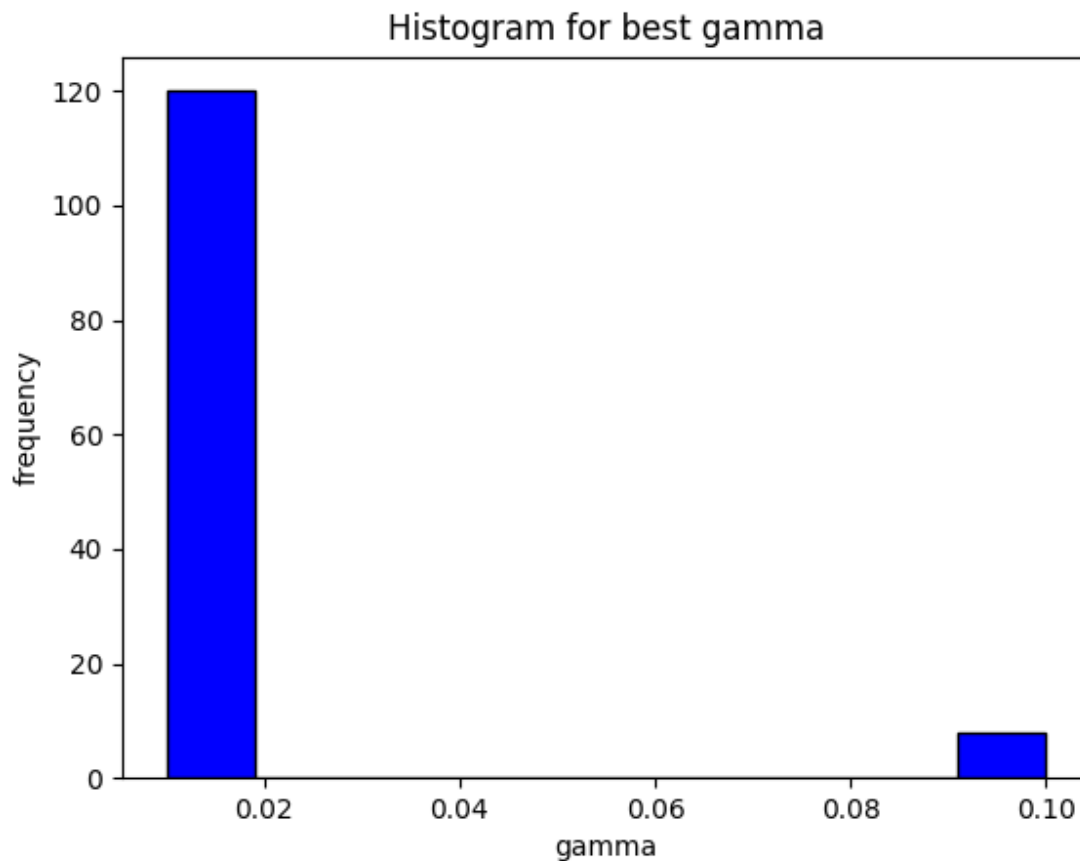
Furthermore, we see that with fixed γ , $\frac{1}{\|w\|}$ decreases quite significantly from $C = 0.1$ to $C = 1$, but from $C = 1$ to $C = 10$, $\frac{1}{\|w\|}$ only decreases a little. This might be because that at $C = 1$ the penalty for violation is already sufficiently large, and there are only few cases of violation. Therefore, further increasing C to 10 does not result in a huge change in the training result, as there's not much room to further decrease the number of violations.

Second, larger γ means that a more complex transform is used, and therefore the model has a higher degree of freedom when fitting the data vectors. Therefore, with a larger γ , the model can find a hyperplane that is further away from the data vectors, and hence the larger margin.

Code Snapshot:

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Histogram:



Findings:

From the histogram above we see that value of the best γ is either 0.01 or 0.1. Moreover, the overwhelming majority is 0.01, with 120 out of 128 values of best γ being 0.01.

A possible explanation to this phenomenon is that when adopting validation, an overfitting model is punished in the model selection step. That is a model that overfits is likely to have high E_{val} and wouldn't be chosen. As we're taught in class, a larger γ in the Gaussian kernel means that there are more effective terms in the transform, and a more complex model is therefore allowed. Therefore, overfitting is more likely to occur, and the model's classifying performance judging by E_{eval} is likely to be worse. As a result, out of the 128 experiments, most of the time the model with the lowest γ factor in the Gaussian kernel performs the best, as it's less prone to overfitting.

Nevertheless, the few cases where $\gamma = 0.1$ performs better than $\gamma = 0.01$ still implies that sometimes

a more complex model does have better classifying performance.
Code Snapshot: